Bayesian Learning

Lecture 05

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Bayesian Formula

- Sum Rule

$$p(X) = \sum_{Y} p(X, Y)$$

- Product Rule

$$p(X, Y) = p(Y|X)p(X)$$

- Bayes theorem

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$$

$$p(X) = \sum_{Y} p(X|Y)p(Y)$$

Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, **53:370-418**

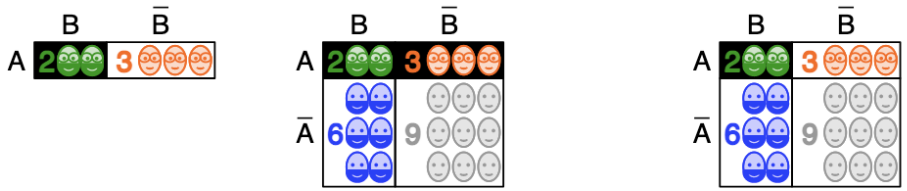
An illustration of Bayes Theorem

From wiki

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$$

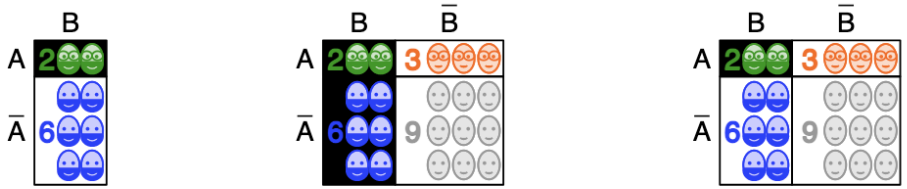
$$p(X) = \sum_{Y} p(X|Y)p(Y)$$

Number of occurences	Beard: B	No beard: $\bar{B}$	sum
Astigmatic: A	2	3	5
Not astigmatic: $\bar{A}$	6	9	15
sum	8	12	20



P(B, given A) · P(A) = P(B|A) · P(A)

$$\frac{2}{\cancel{2+3}} \cdot \frac{\cancel{2+3}}{2+3+6+9} = \frac{2}{2+3+6+9}$$



P(A, given B) · P(B) = P(A|B) · P(B)

$$\frac{2}{\cancel{2+6}} \cdot \frac{\cancel{2+6}}{2+3+6+9} = \frac{2}{2+3+6+9}$$

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$\therefore \quad P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

From distance to probability

distance $\|x-\mu\|^2 \longrightarrow \exp\{-\lambda\|x-\mu\|^2\}$ **likely**

"The closer, the more likely."

Sum or integral to be one

Probability

It is more powerful to consider everything in probability framework!

$$\frac{1}{\sqrt{(2\pi)^k|\boldsymbol{\Sigma}|}}\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

Gaussian distribution with the Mahalanobis distance

$$D_M(x)=\sqrt{(x-\mu)^T\Sigma^{-1}(x-\mu)}.$$

Your first consulting job

- A billionaire asks you a question:
- — He says: I have a coin, if I flip it, what's the probability it will fall with the head up?
- — You say: Please flip it a few times:

- — You say: The probability is : 3/5
- — He says: Why???
- — You say: Because …

Bernoulli Distribution

- Data

$$D = \{x_i\}_{i=1}^{n}, X_i \epsilon \{H, T\}$$

- P(Heads) = θ, P(Tails) = 1- θ,
- Flips are i.i.d.:
  - Independent events
  - Identically distributed according to Bernoulli distribution

<u>Choose θ that maximizes the probability of observed data</u>

Maximum Likelihood Estimation (MLE)

- Choose θ that maximizes the probability of observed data

$$\hat{\theta}_{MLE} = \arg\max_{\theta} P(D|\theta)$$

- MLE of probability of head:

$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}$$

= **3/5** "Frequency of heads"

Number of heads Number of tails

Maximum Likelihood Estimation (MLE)

- Choose θ that maximizes the probability of observed data

$$
\begin{aligned}
\hat{\theta}_{MLE} &= \arg\max_{\theta} P(D|\theta) \\
&= \arg\max_{\theta} \prod_{i=1}^{n} P(X_i|\theta) && \text{Independent draws} \\
&= \arg\max_{\theta} \prod_{i:X_i=H} \theta \prod_{i:X_i=T} (1-\theta) && \text{Identically distributed} \\
&= \arg\max_{\theta} \underbrace{\theta^{\alpha_H}(1-\theta)^{\alpha_T}}_{J(\theta)}
\end{aligned}
$$

Maximum Likelihood Estimation (MLE)

- Choose θ that maximizes the probability of observed data

$$\hat{\theta}_{MLE} = \arg\max_{\theta} P(D|\theta)$$
$$= \arg\max_{\theta} \underbrace{\theta^{\alpha_H}(1-\theta)^{\alpha_T}}_{J(\theta)}$$

$$\frac{\partial J(\theta)}{\partial \theta} = \alpha_H \theta^{\alpha_H - 1}(1-\theta)^{\alpha_T} - \alpha_T \theta^{\alpha_H}(1-\theta)^{\alpha_T - 1}$$
$$= (\alpha_H(1-\theta) - \alpha_T\theta)(\theta^{\alpha_H - 1}(1-\theta)^{\alpha_T - 1})$$

$$\Downarrow$$

$$\alpha_H(1-\theta) - \alpha_T\theta|_{\theta=\hat{\theta}_{MLE}} = 0$$

$$\boxed{\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}}$$

Bayesian Learning

- Bayes rule

$$P(\Theta|X) = \frac{P(X|\Theta)P(\Theta)}{P(X)}$$

$P(X|\Theta)$: likelihood of data X given parameter Θ

$P(\Theta)$: prior distribution over the parameter Θ

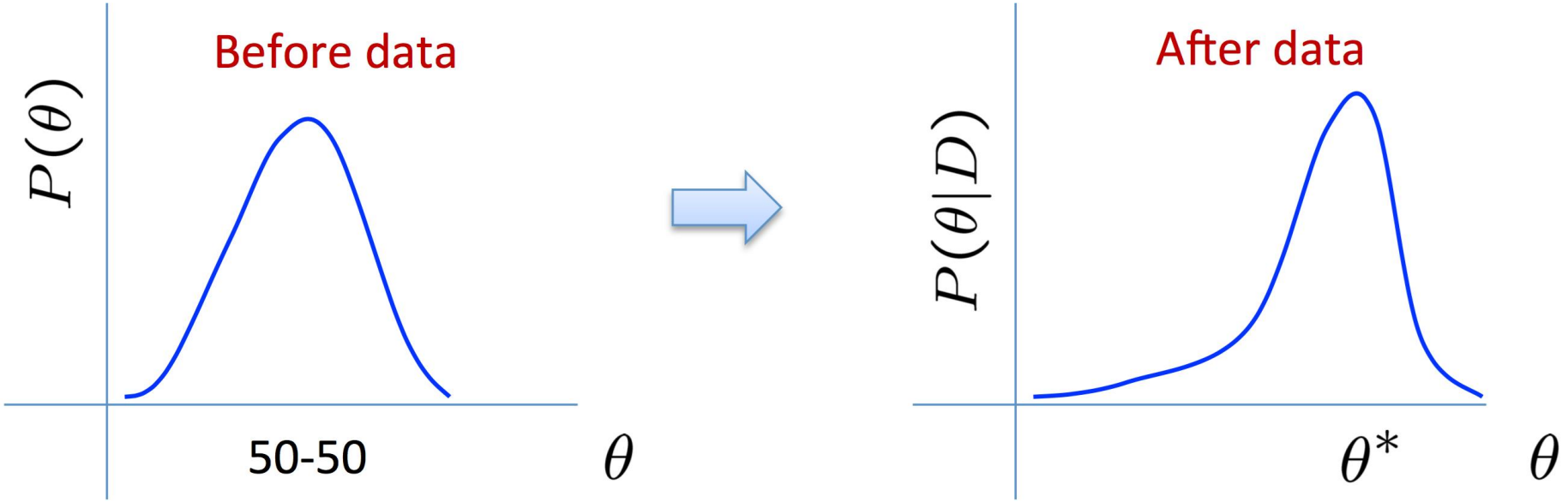
$P(X)$: marginal distribution of data X

Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, **53:370-418**

- Prior distribution
  - Represents expert knowledge
  - Uninformative priors: Uniform distribution
  - Conjugate priors: Closed-form representation of posterior, P(θ) and P(θ|D) have the same form

Back to the example

- Billionaire says: Wait, I know that the coin is “close” to 50-50. What can you do for me now?
- You say: I can learn it the Bayesian way…
- Rather than estimating a single θ, we obtain a distribution over possible values of θ



Conjugate Prior for Coin Flip

- P(θ) and P(θ|D) have the same form

- Likelihood is ~ Binomial

$$P(\mathcal{D} \mid \theta) = \binom{n}{\alpha_H} \theta^{\alpha_H} (1-\theta)^{\alpha_T}$$

- If prior is Beta distribution,

$$P(\theta) = \frac{\theta^{\beta_H - 1}(1-\theta)^{\beta_T - 1}}{B(\beta_H, \beta_T)} \sim Beta(\beta_H, \beta_T)$$

- Then posterior is Beta distribution

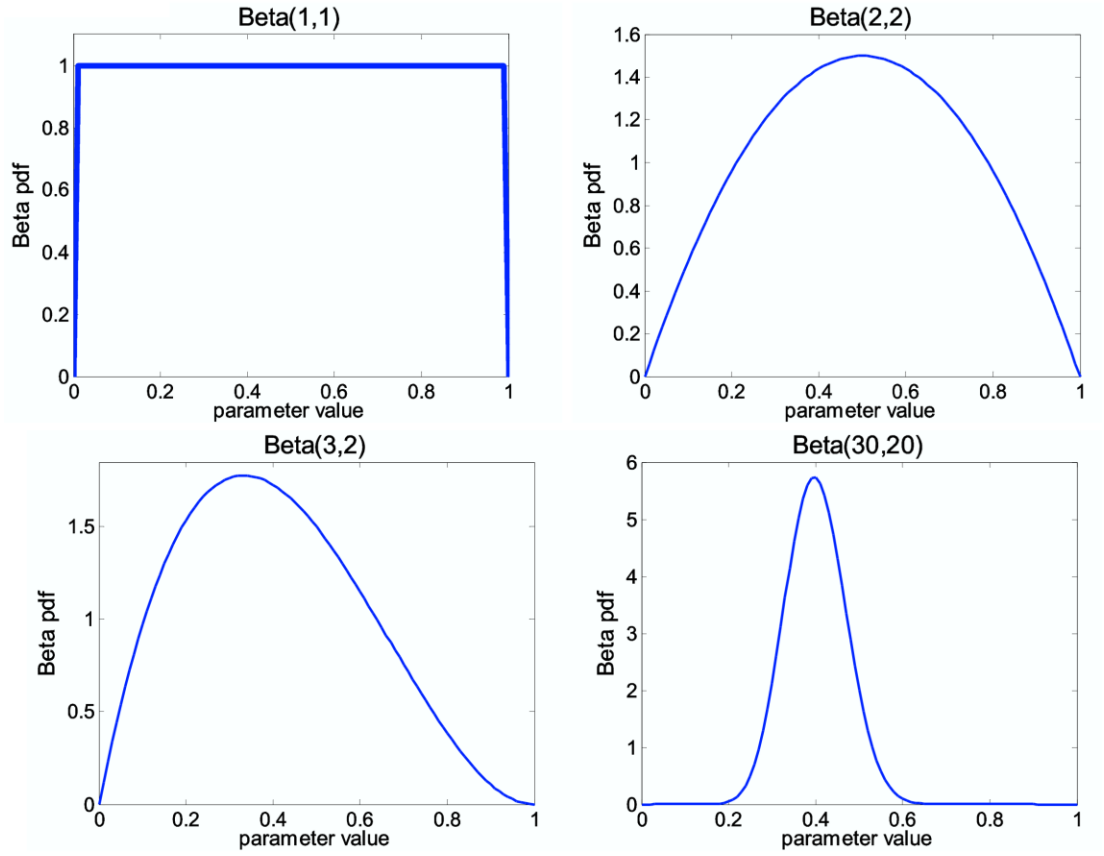
$$P(\theta|D) \sim Beta(\beta_H + \alpha_H, \beta_T + \alpha_T)$$

For Binomial, conjugate prior is Beta distribution

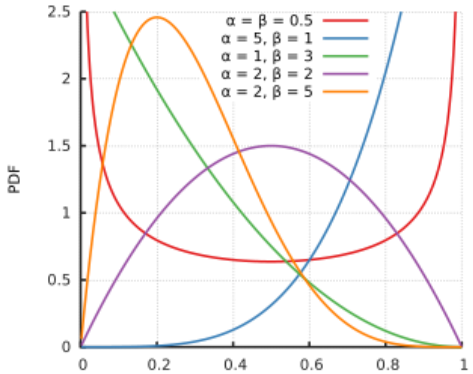
Beta distribution

- More concentrated as values of β_H, β_T increase

$$Beta(\beta_H, \beta_T)$$



Probability density function



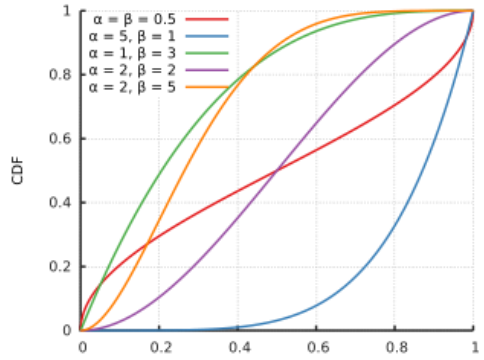
$$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}$$

$$\mathrm{E}[X] = \frac{\alpha}{\alpha+\beta}$$

$$\mathrm{B}(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$\mathrm{var}[X] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Cumulative distribution function



mode

$$\frac{\alpha-1}{\alpha+\beta-2} \text{ for } \alpha, \beta > 1$$

$I_x(\alpha,\beta)$

(the regularized incomplete beta function)

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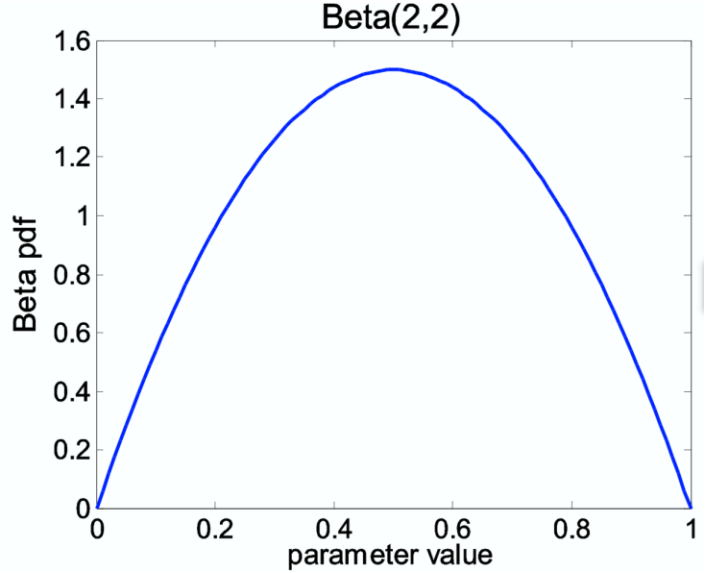
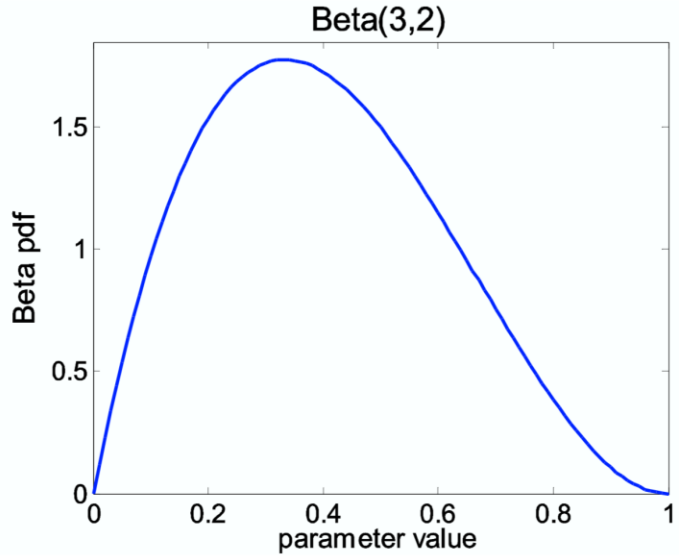
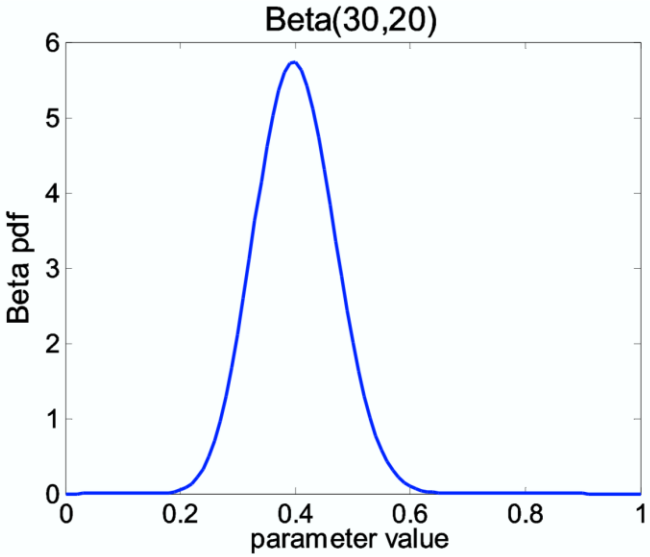
Beta Conjugate Prior

$$P(\theta) \sim Beta(\beta_H, \beta_T)$$

$$P(\theta|D) \sim Beta(\beta_H + \alpha_H, \beta_T + \alpha_T)$$







$As\ n = \alpha_H + \alpha_T\ increases$

As we get more samples, effect of prior is “washed out”

Conjugate Prior for Dice Roll (*6 outcomes instead of 2*)

- P(θ) and P(θ|D) have the same form

- Likelihood is ~ Multinomial ($\theta = \{\theta_1, \theta_2, \ldots, \theta_k\}$)

$$P(\mathcal{D} \mid \theta) = \frac{n!}{\alpha_1!\alpha_2!\ldots\alpha_k!}\theta_1^{\alpha_1}\theta_2^{\alpha_2}\ldots\theta_k^{\alpha_k} \qquad \sum_{i=1}^{k}\alpha_i = n \qquad \sum_{i=1}^{k}\theta_i = 1$$

- If prior is Dirichlet distribution,

$$P(\theta) = \frac{\prod_{i=1}^{k}\theta_i^{\beta_i - 1}}{B(\beta_1, \ldots, \beta_k)} \sim Dirichlet(\beta_1, \ldots, \beta_k)$$

- Then posterior is Dirichlet distribution

$$P(\theta|D) \sim Dirichlet(\beta_1 + \alpha_1, \ldots, \beta_k + \alpha_k)$$

For Multinomial, conjugate prior is Dirichlet distribution

Maximum A Posterior (MAP) Estimation

- Choose θ that maximizes a posterior probability

$$\hat{\theta}_{MAP} = \arg\max_{\theta} P(\theta|D)$$
$$= \arg\max_{\theta} P(D|\theta)P(\theta)$$

- MAP estimate of probability of head:

$$P(\theta|D) \sim Beta(\beta_H + \alpha_H, \beta_T + \alpha_T)$$

$$\hat{\theta}_{MAP} = \frac{\alpha_H + \beta_H - 1}{\alpha_H + \beta_H + \alpha_T + \beta_T - 2}$$

Mode of Beta distribution

MLE vs. MAP

- Maximum likelihood estimation (MLE)
  - Choose value that maximizes the probability of observed data

$$\hat{\theta}_{MLE} = \arg\max_{\theta} P(D|\theta)$$

- Maximum a posteriori (MAP) estimation
  - Choose value that is most probable given observed data and prior belief

$$\begin{aligned}\hat{\theta}_{MAP} &= \arg\max_{\theta} P(\theta|D) \\ &= \arg\max_{\theta} p(D|\theta)P(\theta)\end{aligned}$$

MLE vs. MAP

- What if we toss the coin too few times?

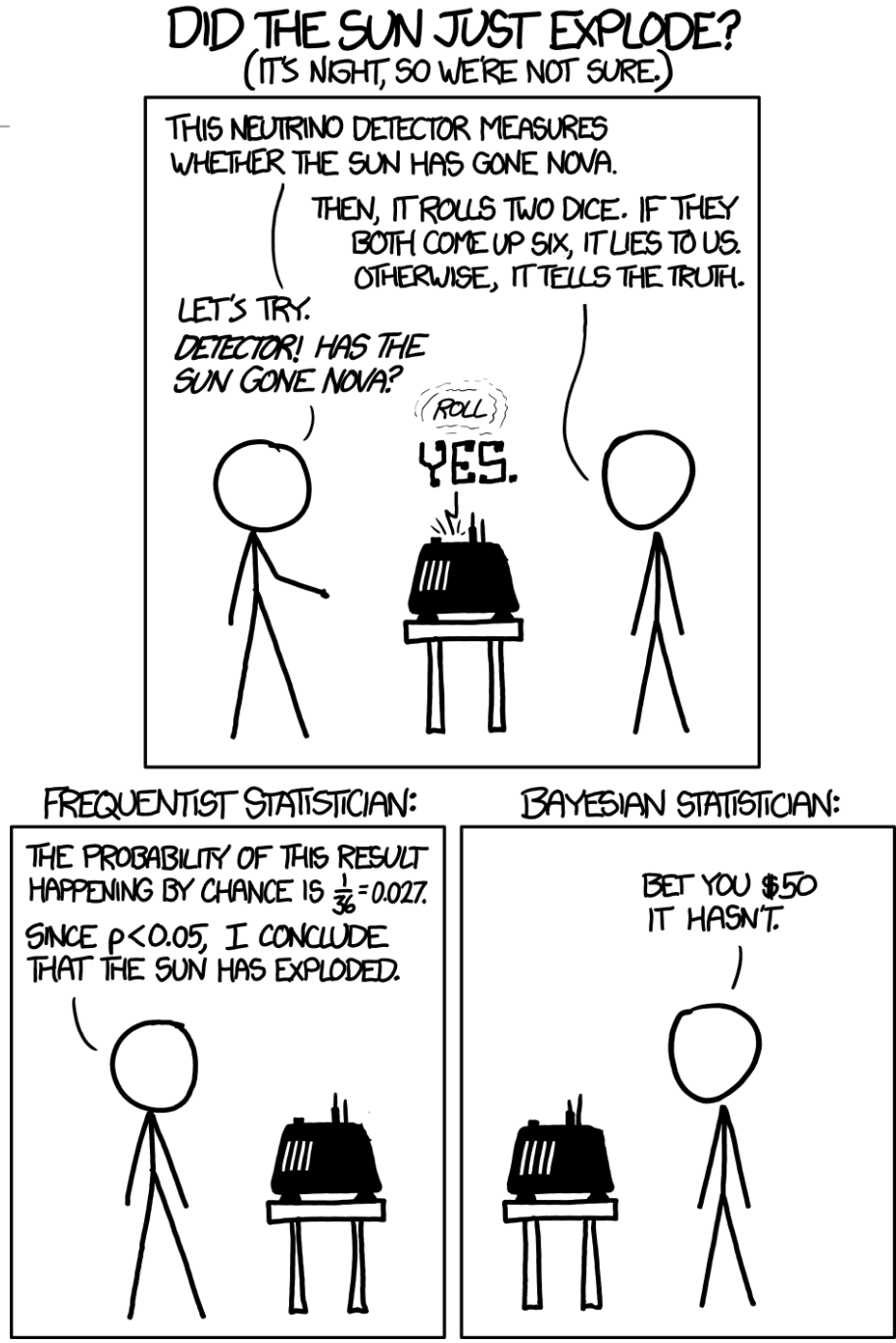
$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}$$

- You say: Probability next toss is a head = 0
- Billionaire says: You're fired!

- Beta prior equivalent to extra coin flips
- As n → infinity, prior is "forgotten"
- But, for small sample size, prior is important!

$$\hat{\theta}_{MAP} = \frac{\alpha_H + \beta_H - 1}{\alpha_H + \beta_H + \alpha_T + \beta_T - 2}$$

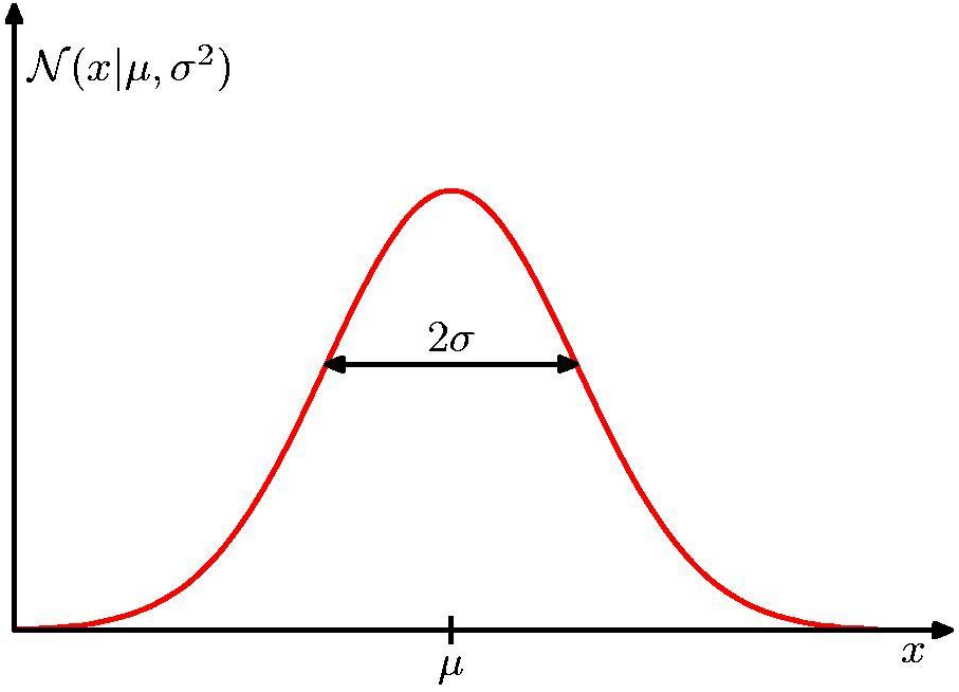
Bayesians vs. Frequentists



What about continuous variables?

- The Gaussian Distribution

$$\mathcal{N}\left(x|\mu,\sigma^2\right) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$



$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) x \, \mathrm{d}x = \mu$$

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) x^2 \, \mathrm{d}x = \mu^2 + \sigma^2$$

$$\mathrm{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = \sigma^2$$

$$\mathcal{N}(x|\mu,\sigma^2) > 0 \qquad \int_{-\infty}^{\infty} \mathcal{N}\left(x|\mu,\sigma^2\right) \, \mathrm{d}x = 1$$

MLE for Gaussian mean and variance

- Choose $\theta = (\mu, \sigma^2)$ that maximizes the probability of observed data

$$\hat{\theta}_{MLE} = \underset{\theta}{argmax}\, P(D|\theta)$$

$$= \underset{\theta}{argmax} \prod_{i=1}^{n} P(x_i|\theta)$$

$$= \underset{\theta}{argmax} \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

Independent draws

Identically distributed

$$\ln p(D|\theta) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{n}{2}\ln\sigma^2 - \frac{n}{2}\ln(2\pi)$$

MLE for Gaussian mean and variance

$$\ln p(D|\theta) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{n}{2}\ln\sigma^2 - \frac{n}{2}\ln(2\pi)$$

$$\frac{\partial \ln p(D|\mu,\sigma^2)}{\partial \mu} = 0 \quad \Rightarrow \quad \hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\frac{\partial \ln p(D|\mu,\sigma^2)}{\partial \sigma^2} = 0 \quad \Rightarrow \quad \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2$$

- Note: MLE for the variance of a Gaussian is biased
  - Unbiased variance estimator:

$$\hat{\sigma}^2_{unbiased} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\hat{\mu})^2$$

MAP for Gaussian mean and variance

- Conjugate priors
  - Mean: Gaussian prior
  - Variance: Wishart Distribution

$$p(\mu) = \frac{1}{\sqrt{2\pi}\beta} e^{-\frac{(\mu-\nu)^2}{2\beta^2}}$$

- Posterior for mean:

$$p(\mu|D) = \frac{p(D|\mu)p(\mu)}{p(D)} = \frac{(\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}) \frac{1}{\sqrt{2\pi}\beta} e^{-\frac{(\mu-\nu)^2}{2\beta^2}}}{p(D)}$$

- Take the log of both side, and take the derivative w.r.t. μ. We have:

$$\frac{\partial \ln p(\mu|D)}{\partial \mu} = (\sum_{i=1}^{n} \frac{x_i - \mu}{\sigma^2}) - \frac{\mu - \nu}{\beta^2} = 0 \quad \Rightarrow \quad \hat{\mu} = \frac{\sigma^2 \nu + \beta^2 \sum_{i=1}^{n} x_i}{\sigma^2 + N\beta^2}$$

MLE vs. MAP

- As N→∞, $\hat{\mu}_{MAP} \rightarrow \hat{\mu}_{MLE}$

$$\hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\hat{\mu}_{MAP} = \frac{\sigma^2 \nu}{\sigma^2 + \mathrm{n}\beta^2} + \frac{\frac{1}{\mathrm{n}}\sum_{i=1}^{n} x_i}{1 + \frac{\sigma^2}{\mathrm{n}\beta^2}}$$

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